

Homework (Habanero)

- Q1.** What is the value of the digit 7 in the number 74521? (A) 320
(A) $7 \cdot 10^4$ (B) 330
(B) $7 \cdot 10^3$ (C) 324
(C) $7 \cdot 10^2$ (D) 324.9
(D) $7 \cdot 10^1$ (E) 325
(E) $7 \cdot 10^0$
- Q2.** Round 456.789 to the nearest 10 (A) 450
(A) 450 (B) 460
(B) 460 (C) 457
(C) 457 (D) 456.8
(D) 456.8 (E) 456.79
(E) 456.79
- Q3.** A patient needs 0.0456 liters of medicine. Write this in expanded form (A) $4 \cdot 10^{-2} + 5 \cdot 10^{-3} + 6 \cdot 10^{-4}$
(A) $4 \cdot 10^{-2} + 5 \cdot 10^{-3} + 6 \cdot 10^{-4}$ (B) $4 \cdot 10^{-1} + 5 \cdot 10^{-2} + 6 \cdot 10^{-3}$
(B) $4 \cdot 10^{-1} + 5 \cdot 10^{-2} + 6 \cdot 10^{-3}$ (C) $4 \cdot 10^{-2} + 5 \cdot 10^{-3} + 6 \cdot 10^{-4}$
(C) $4 \cdot 10^{-2} + 5 \cdot 10^{-3} + 6 \cdot 10^{-4}$ (D) $45 \cdot 10^{-3} + 6 \cdot 10^{-4}$
(D) $45 \cdot 10^{-3} + 6 \cdot 10^{-4}$ (E) $46 \cdot 10^{-3}$
(E) $46 \cdot 10^{-3}$
- Q4.** The population of a cell grows to 34,567. Write this number in standard form (A) $3.4567 \cdot 10^4$
(A) $3.4567 \cdot 10^4$ (B) $3.4567 \cdot 10^5$
(B) $3.4567 \cdot 10^5$ (C) $3.4567 \cdot 10^3$
(C) $3.4567 \cdot 10^3$ (D) 345.67
(D) 345.67 (E) 34567
(E) 34567
- Q5.** A doctor prescribes 324.9 milligrams of a certain medication to a patient. Round to the nearest 10 milligrams
- Q6.** A lab contains 0.000543 liters of a chemical solution. Write this number in expanded form (A) $5 \cdot 10^{-4} + 4 \cdot 10^{-5} + 3 \cdot 10^{-6}$
(A) $5 \cdot 10^{-4} + 4 \cdot 10^{-5} + 3 \cdot 10^{-6}$ (B) $5 \cdot 10^{-1} + 4 \cdot 10^{-2} + 3 \cdot 10^{-3}$
(B) $5 \cdot 10^{-1} + 4 \cdot 10^{-2} + 3 \cdot 10^{-3}$ (C) $5 \cdot 10^{-3} + 4 \cdot 10^{-4} + 3 \cdot 10^{-5}$
(C) $5 \cdot 10^{-3} + 4 \cdot 10^{-4} + 3 \cdot 10^{-5}$ (D) $5 \cdot 10^{-4} + 4 \cdot 10^{-5} + 3 \cdot 10^{-6}$
(D) $5 \cdot 10^{-4} + 4 \cdot 10^{-5} + 3 \cdot 10^{-6}$ (E) $543 \cdot 10^{-6}$
(E) $543 \cdot 10^{-6}$
- Q7.** The length of a bacterium is 0.0025 millimeters. Write this in standard form (A) $2.5 \cdot 10^{-3}$
(A) $2.5 \cdot 10^{-3}$ (B) $2.5 \cdot 10^{-2}$
(B) $2.5 \cdot 10^{-2}$ (C) $2.5 \cdot 10^{-4}$
(C) $2.5 \cdot 10^{-4}$ (D) $2.5 \cdot 10^{-1}$
(D) $2.5 \cdot 10^{-1}$ (E) $2.5 \cdot 10^1$
(E) $2.5 \cdot 10^1$
- Q8.** A certain medical treatment costs 2,456.78
2,456.78
. Round the cost to the nearest 100 (A) 2,500
(A) 2,500 (B) 2,400
(B) 2,400 (C) 2,456.80
(C) 2,456.80 (D) 2,456.78
(D) 2,456.78 (E) 2,457
(E) 2,457

Open Ended Questions (FRQ)

- Q9.** A hospital has 5,678,923 patients. Write this number in expanded form and round it to the nearest 10,000. Provide your calculations.
- Q10.** A doctor prescribes a medicine with a dosage of 45.6789 milligrams per kilogram of body weight. Write this number in standard form and round it to the nearest 0.01 milligrams. Calculate the total dosage for a 75 kilogram patient.

Chef's Tips

- Remind students to use the correct number of significant figures when rounding numbers
- Encourage students to use expanded and standard form when working with large or small numbers
- Emphasize the importance of precise calculations when working with medical dosages and patient information